Face Mask Detection Project

In [ ]:

In [1]:

```
#Import the dataset:
import kagglehub

# Download latest version
path = kagglehub.dataset_download("omkargurav/face-mask-dataset")

print("Path to dataset files:", path)
```

```
Downloading to C:\Users\Lenovo\.cache\kagglehub\datasets\omkargurav\face-mask-dataset\1.a
rchive...
```

```
100%|██████████████████████████████████████████████████████████████████████████| 163
M/163M [00:52<00:00, 3.28MB/s]
```

```
Extracting files...
```

```
Path to dataset files: C:\Users\Lenovo\.cache\kagglehub\datasets\omkargurav\face-mask-dat
aset\versions\1
```

In [ ]:

In [1]:

```
#Import the dependencies:
import os
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import cv2
from PIL import Image
from sklearn.model_selection import train_test_split
```

In [3]:

```
with_mask_files = os.listdir("data/with_mask")
without_mask_files = os.listdir("data/without_mask")

print("Number of:")
print("\t\t With Mask Images = ",len(with_mask_files))
print("\t\t Without Mask Images = ",len(without_mask_files))
```

```
Number of:
   With Mask Images =  3725
   Without Mask Images =  3828
```

In [4]:

```
#Create labels for both classes of images:
### With mask = 1
### Without mask = 0


with_mask_labels = [1]*len(with_mask_files)
without_mask_labels = [0]*len(without_mask_files)
```

In [5]:

```
#Labels to first 5 images with masks:
with_mask_labels[:5]
```

Out[5]:

```
[1, 1, 1, 1, 1]
```

In [8]:

```
#Labels to last 5 images without masks:
without_mask_labels[-5:]
```

Out[8]:

```
[0, 0, 0, 0, 0]
```

In [10]:

```
print("Number of:")
print("\t\t With Mask Images = ",len(with_mask_labels))
print("\t\t Without Mask Images = ",len(without_mask_labels))


labels = with_mask_labels + without_mask_labels
print(len(labels))
```

```
Number of:
   With Mask Images =  3725
   Without Mask Images =  3828
7553
```

In [ ]:

```

```

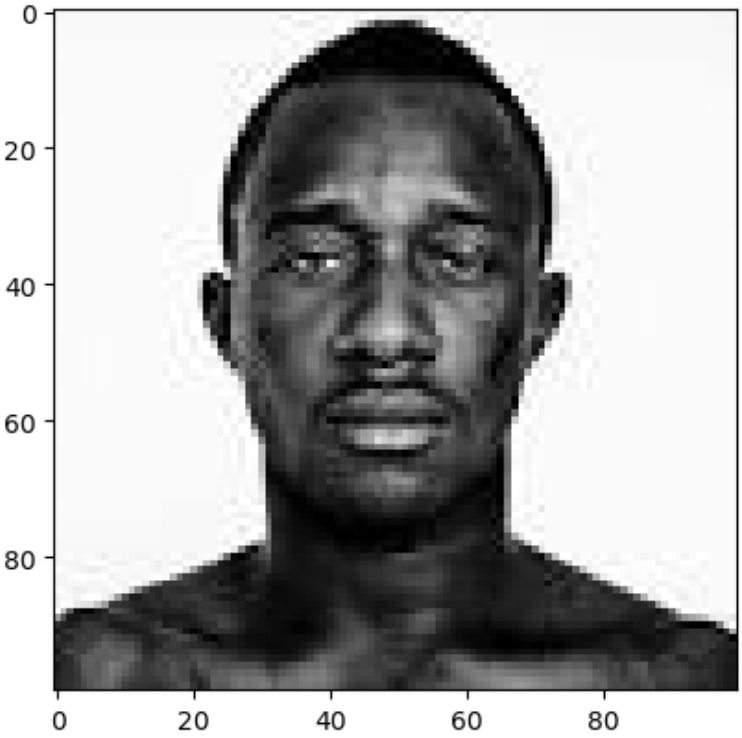
In [ ]:

```

```

Displaying the images:

In [14]:

```
info = {"With Mask":"data/with_mask/with_mask_1.jpg","Without Mask":"data/without_mask/without_mask_1.jpg"}

for i , j in info.items():
    img = mpimg.imread(j)
    imgplot = plt.imshow(img)
    plt.show()
```



CONCLUSION AT THIS POINT] Image dimensions are not uniform.

In [ ]:

```

```

In [ ]:

```

```

Image Preprocessing:

1. **Resize the images**
2. **Convert the images to numpy arrays**

In [16]:

```
#Preprocessing for with_mask_images:
with_mask_path = "data/with_mask/"

data = []

for img_file in with_mask_files:
    image = Image.open(with_mask_path + img_file)
    image = image.resize((128,128))
    image = image.convert("RGB")
    image = np.array(image)
    data.append(image)




#Preprocessing for without_mask_images:
without_mask_path = "data/without_mask/"

for img_file in without_mask_files:
```

```
    image = Image.open(without_mask_path + img_file)
    image = image.resize((128,128))
    image = image.convert("RGB")
    image = np.array(image)
    data.append(image)
```

```
C:\Users\Lenovo\anaconda3\Lib\site-packages\PIL\Image.py:1045: UserWarning: Palette image
s with Transparency expressed in bytes should be converted to RGBA images
  warnings.warn(
```

In [17]:

```
len(data)
```

Out[17]:

```
7553
```

In [18]:

```
type(data)
```

Out[18]:

```
list
```

In [19]:

```
#View the first image (Converted into an numpy array):
data[0]
```

Out[19]:

```
array([[[50, 58, 31],
        [46, 54, 34],
        [56, 59, 42],
        ...,
        [26, 32, 25],
        [29, 33, 18],
        [34, 40, 16]],

       [[47, 55, 30],
        [49, 57, 37],
        [59, 63, 40],
        ...,
        [18, 21, 11],
        [38, 42, 31],
        [31, 41, 22]],

       [[52, 58, 37],
        [50, 59, 38],
        [59, 64, 36],
        ...,
        [45, 44, 30],
        [29, 35, 28],
        [20, 30, 17]],

       ...,

       [[63, 72, 37],
        [66, 76, 41],
        [74, 84, 50],
        ...,
        [ 5,  4,  0],
        [20, 19, 15],
        [17, 17, 12]],

       [[59, 71, 34],
        [60, 70, 39],
        [68, 76, 50],
        ...,
        [16, 15, 12],
        [15, 13, 10],
```

```
        [13,  9,  7]],

       [[53, 69, 29],
        [54, 64, 37],
        [57, 63, 43],
        ...,
        [18, 18, 16],
        [16, 13, 12],
        [20, 12, 13]]], shape=(128, 128, 3), dtype=uint8)
```

In [20]:

```
type(data[0])
```

Out[20]:

```
numpy.ndarray
```

In [21]:

```
data[0].shape #height, width and color channel (respectively) of the first image
```

Out[21]:

```
(128, 128, 3)
```

In [ ]:

```

```

In [ ]:

```

```

Convert both the data-list and labels into numpy arrays:

In [23]:

```
X = np.array(data)
Y = np.array(labels)

print(type(X))
print(type(Y))
```

```
<class 'numpy.ndarray'>
<class 'numpy.ndarray'>
```

In [26]:

```
print(X.shape) #represents (no. of images (arrays), height of each image individually, wi
dth of each image individually, color channel)
print(Y.shape) #no. of labels
```

```
(7553, 128, 128, 3)
(7553,)
```

In [28]:

```
#Split your data into training and testing data:
X_train , X_test, Y_train, Y_test = train_test_split(X,Y,test_size=0.2, random_state=42)
```

In [32]:

```
#Data before scaling:
X_train[0]
```

Out[32]:

```
array([[[ 40,  17,  13],
        [ 42,  18,  14],
        [ 46,  20,  16],
```

```
        ...,
        [ 45,  29,  30],
        [ 44,  29,  32],
        [ 44,  29,  32]],

       [[ 44,  18,  15],
        [ 47,  21,  17],
        [ 54,  25,  21],
        ...,
        [ 45,  30,  31],
        [ 44,  29,  32],
        [ 44,  29,  32]],

       [[ 53,  23,  18],
        [ 58,  27,  22],
        [ 69,  36,  30],
        ...,
        [ 44,  29,  30],
        [ 43,  28,  31],
        [ 43,  28,  31]],

       ...,

       [[138, 186, 165],
        [135, 186, 164],
        [127, 182, 161],
        ...,
        [177, 120, 111],
        [179, 122, 112],
        [181, 123, 114]],

       [[153, 189, 171],
        [150, 189, 170],
        [145, 188, 168],
        ...,
        [176, 119, 108],
        [178, 121, 109],
        [178, 121, 110]],

       [[158, 196, 177],
        [155, 196, 178],
        [151, 196, 176],
        ...,
        [174, 117, 106],
        [175, 118, 107],
        [175, 118, 107]]], shape=(128, 128, 3), dtype=uint8)
```

In [33]:

```
#Scale the data (for making the CNN neural network work better):
X_train_scaled = X_train/255
X_test_scaled = X_test/255
```

In [36]:

```
#Data after scaling:
X_train_scaled[0]
```

Out[36]:

```
array([[[0.15686275, 0.06666667, 0.05098039],
        [0.16470588, 0.07058824, 0.05490196],
        [0.18039216, 0.07843137, 0.0627451 ],
        ...,
        [0.17647059, 0.11372549, 0.11764706],
        [0.17254902, 0.11372549, 0.1254902 ],
        [0.17254902, 0.11372549, 0.1254902 ]],

       [[0.17254902, 0.07058824, 0.05882353],
        [0.18431373, 0.08235294, 0.06666667],
        [0.21176471, 0.09803922, 0.08235294],
        ...,
        [0.17647059, 0.11764706, 0.12156863]
```

```
        [0.17647059, 0.11764706, 0.12156863],
        [0.17254902, 0.11372549, 0.1254902 ],
        [0.17254902, 0.11372549, 0.1254902 ]],

       [[0.20784314, 0.09019608, 0.07058824],
        [0.22745098, 0.10588235, 0.08627451],
        [0.27058824, 0.14117647, 0.11764706],
        ...,
        [0.17254902, 0.11372549, 0.11764706],
        [0.16862745, 0.10980392, 0.12156863],
        [0.16862745, 0.10980392, 0.12156863]],

       ...,

       [[0.54117647, 0.72941176, 0.64705882],
        [0.52941176, 0.72941176, 0.64313725],
        [0.49803922, 0.71372549, 0.63137255],
        ...,
        [0.69411765, 0.47058824, 0.43529412],
        [0.70196078, 0.47843137, 0.43921569],
        [0.70980392, 0.48235294, 0.44705882]],

       [[0.6       , 0.74117647, 0.67058824],
        [0.58823529, 0.74117647, 0.66666667],
        [0.56862745, 0.7372549 , 0.65882353],
        ...,
        [0.69019608, 0.46666667, 0.42352941],
        [0.69803922, 0.4745098 , 0.42745098],
        [0.69803922, 0.4745098 , 0.43137255]],

       [[0.61960784, 0.76862745, 0.69411765],
        [0.60784314, 0.76862745, 0.69803922],
        [0.59215686, 0.76862745, 0.69019608],
        ...,
        [0.68235294, 0.45882353, 0.41568627],
        [0.68627451, 0.4627451 , 0.41960784],
        [0.68627451, 0.4627451 , 0.41960784]]], shape=(128, 128, 3))
```

In [ ]:

```

```

In [ ]:

```

```

Building a CNN (Convolutional Neural Network):

In [37]:

```
import tensorflow as tf
from tensorflow import keras
```

In [38]:

```
num_of_classes = 2

model = keras.Sequential()

model.add(keras.layers.Conv2D(32, kernel_size=(3,3), activation='relu', input_shape=(128
,128,3))) #we used 3 as the last parameter since we also have colored images and not only
Gray-scale images
model.add(keras.layers.MaxPooling2D(pool_size=(2,2)))

model.add(keras.layers.Conv2D(64, kernel_size=(3,3), activation='relu'))
model.add(keras.layers.MaxPooling2D(pool_size=(2,2)))

model.add(keras.layers.Flatten())
```

```
#Create a dense layer:
model.add(keras.layers.Dense(128, activation='relu'))
#Prevent model being over-fitted:
model.add(keras.layers.Dropout(0.5))



#Create an another dense layer:
model.add(keras.layers.Dense(64, activation='relu'))
#Prevent model being over-fitted:
model.add(keras.layers.Dropout(0.5))


#Build the output layer:
model.add(keras.layers.Dense(num_of_classes,activation='sigmoid'))
```

```
C:\Users\Lenovo\anaconda3\Lib\site-packages\keras\src\layers\convolutional\base_conv.py:1
13: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in the model
instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

In [39]:

```
#Compile the neural network:
model.compile(optimizer='adam',
              loss="sparse_categorical_crossentropy",
             metrics=["acc"])
```

In [41]:

```
#Train the neural network:

history = model.fit(X_train_scaled,Y_train,validation_split=0.1,epochs=5) #Low epochs si
nce dont have large data and giving a large value will result in the model getting overfi
tted.
```

```
Epoch 1/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 51s 277ms/step - acc: 0.7901 - loss: 0.4720 - val_acc: 0.900
8 - val_loss: 0.2265
Epoch 2/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 46s 270ms/step - acc: 0.8858 - loss: 0.2947 - val_acc: 0.925
6 - val_loss: 0.1744
Epoch 3/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 47s 277ms/step - acc: 0.9068 - loss: 0.2310 - val_acc: 0.937
2 - val_loss: 0.1732
Epoch 4/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 81s 270ms/step - acc: 0.9215 - loss: 0.2024 - val_acc: 0.943
8 - val_loss: 0.1488
Epoch 5/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 47s 276ms/step - acc: 0.9343 - loss: 0.1749 - val_acc: 0.938
8 - val_loss: 0.1354
```

Model Evaluation:

In [43]:

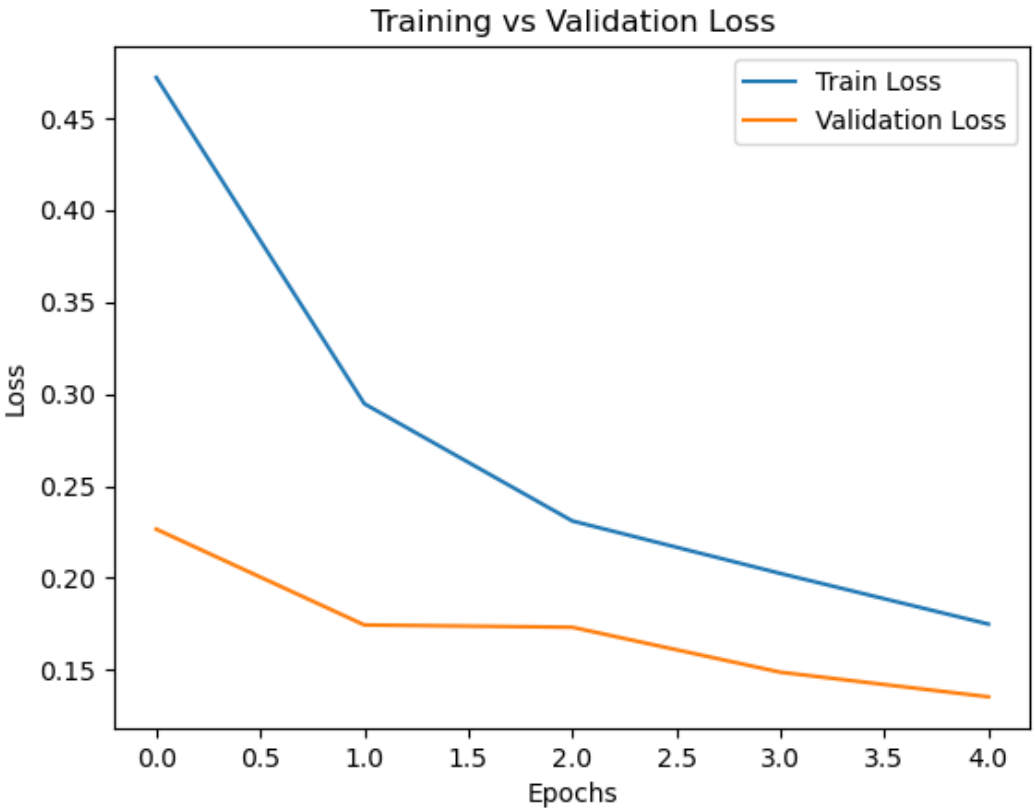
```
loss , accuracy = model.evaluate(X_test_scaled , Y_test)
print("Test Accuracy = ",accuracy*100,"%")
```

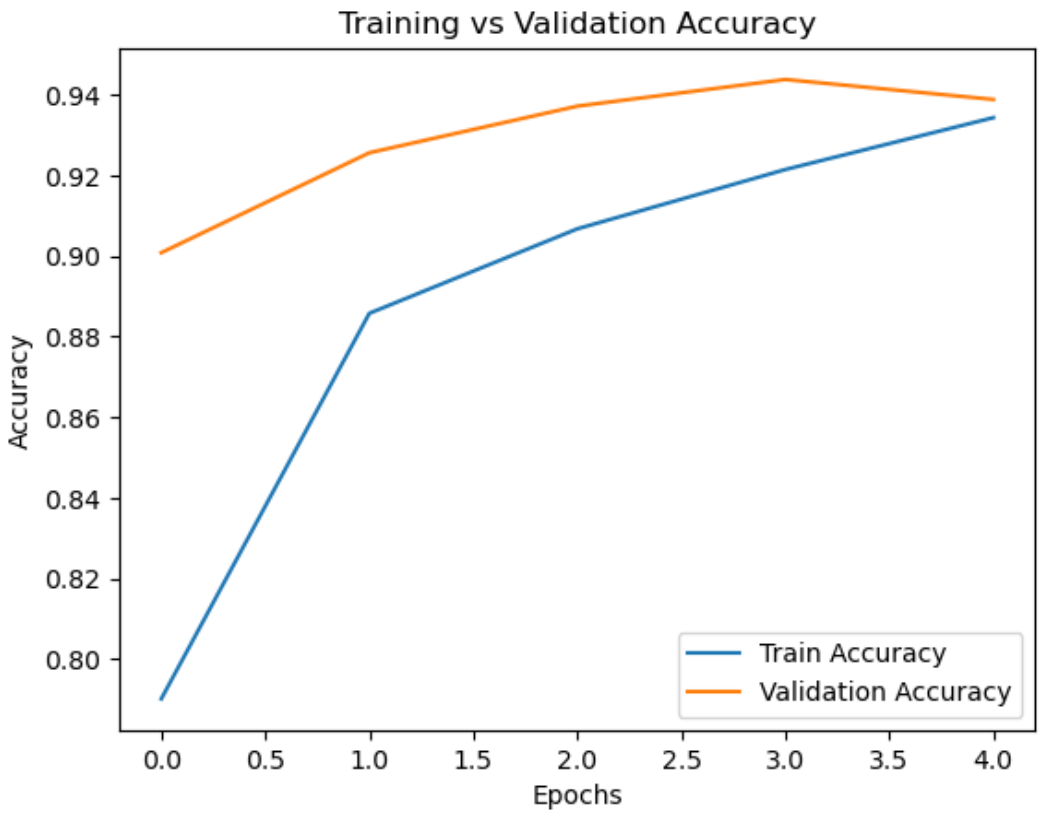
```
48/48 ━━━━━━━━━━━━━━━━━━━━ 4s 75ms/step - acc: 0.9226 - loss: 0.2055
Test Accuracy =  92.25678443908691 %
```

In [47]:

```
h = history

# Loss graph
plt.plot(h.history['loss'], label='Train Loss')
plt.plot(h.history['val_loss'], label='Validation Loss')
```

```
plt.title("Training vs Validation Loss")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
plt.show()

# Accuracy graph
plt.plot(h.history['acc'], label='Train Accuracy')
plt.plot(h.history['val_acc'], label='Validation Accuracy')
plt.title("Training vs Validation Accuracy")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```





In [ ]:

Make Predictions using the model:

In [50]:

```
input_image_path = input('Path of the image to be predicted: ')
input_image = cv2.imread(input_image_path)
plt.imshow(input_image)

input_image_resized = cv2.resize(input_image, (128,128))
input_image_scaled = input_image_resized / 255
input_image_reshaped = np.reshape(input_image_scaled, [1,128,128,3]) #Make prediction fo
r one image with dimension as (128,128,3)

input_prediction = model.predict(input_image_reshaped)
print(input_prediction)
input_pred_label = np.argmax(input_prediction)
print(input_pred_label)


if input_pred_label == 1:
    print("The person is wearing a mask")
else:
    print("The person is not wearing mask")
```

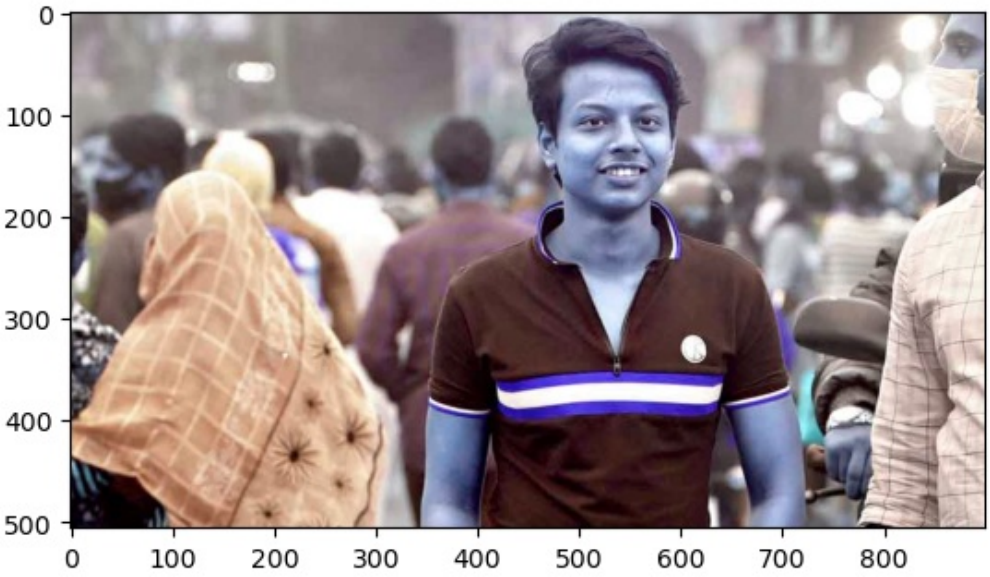
```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 120ms/step
[[0.54502964 0.81090623]]
1
The person is wearing a mask
```

In [51]:

```
input_image_path = input('Path of the image to be predicted: ')
input_image = cv2.imread(input_image_path)
plt.imshow(input_image)

input_image_resized = cv2.resize(input_image, (128,128))
input_image_scaled = input_image_resized / 255
input_image_reshaped = np.reshape(input_image_scaled, [1,128,128,3]) #Make prediction fo
r one image with dimension as (128,128,3)

input_prediction = model.predict(input_image_reshaped)
print(input_prediction)
input_pred_label = np.argmax(input_prediction)
print(input_pred_label)
```

```
if input_pred_label == 1:
    print("The person is wearing a mask")
else:
    print("The person is not wearing mask")
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 74ms/step
[[0.59954166 0.399003  ]]
0
The person is not wearing mask
```



In [ ]:

Deploy the model:

In [52]:

```
# Save
model.save("model.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.savin
g.save_model(model)`. This file format is considered legacy. We recommend using instead t
he native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(m
odel, 'my_model.keras')`.
```

In [ ]:

Test the exported model on some new image:

In [53]:

```
# Load the model and make predictions
from tensorflow.keras.models import load_model
cnn_model = load_model("model.h5")


input_image_path = input('Path of the image to be predicted: ')
input_image = cv2.imread(input_image_path)
plt.imshow(input_image)

input_image_resized = cv2.resize(input_image, (128,128))
input_image_scaled = input_image_resized / 255
input_image_reshaped = np.reshape(input_image_scaled, [1,128,128,3]) #Make prediction fo
```

```
r one image with dimension as (128,128,3)

input_prediction = cnn_model.predict(input_image_reshaped)
print(input_prediction)
input_pred_label = np.argmax(input_prediction)
print(input_pred_label)


if input_pred_label == 1:
    print("The person is wearing a mask")
else:
    print("The person is not wearing mask")
```

```
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `m
odel.compile_metrics` will be empty until you train or evaluate the model.
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 147ms/step
[[0.54064167 0.47327387]]
0
The person is not wearing mask
```

In [54]:

```
# Load the model and make predictions
from tensorflow.keras.models import load_model
cnn_model = load_model("model.h5")


input_image_path = input('Path of the image to be predicted: ')
input_image = cv2.imread(input_image_path)
plt.imshow(input_image)

input_image_resized = cv2.resize(input_image, (128,128))
input_image_scaled = input_image_resized / 255
input_image_reshaped = np.reshape(input_image_scaled, [1,128,128,3]) #Make prediction fo
r one image with dimension as (128,128,3)

input_prediction = cnn_model.predict(input_image_reshaped)
print(input_prediction)
input_pred_label = np.argmax(input_prediction)
print(input_pred_label)


if input_pred_label == 1:
    print("The person is wearing a mask")
else:
    print("The person is not wearing mask")
```

```
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `m
odel.compile_metrics` will be empty until you train or evaluate the model.
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 146ms/step
```

```
[[0.5782794  0.33195263]]
0
The person is not wearing mask
```

In [ ]:

In [ ]:

CONCLUSION AT THIS POINT] The CNN model made incorrect prediction for the third image and has an accuracy of 94% .

In [ ]:

In [ ]:

CONCLUSION]

This project was demonstrating the use of CNN model to detect whether a person is wearing a face-mask or not.

The accuracy of this model is 94% and it was trained for 5 epochs, and backpropagation was performed 850 times (since 170 * 5) per epoch.

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

```

```

Project made by:

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Date of project completion: 30 April 2026

In [ ]:

```

```